

Product Requirements Document (PRD): Cross-Platform Book Tracker

Project Lead: Abhijeet Singh Rathore **Target Ecosystem:** Android (Mobile/Tablet), Wear OS, Desktop (PWA) **Tech Stack:** Kotlin, Jetpack Compose (MD3), Firebase (Firestore/Auth), Web (Progressive Web App)

1. Core Infrastructure & Table Stakes

- **The Big 3 Status Shelves:** Standard "Currently Reading," "Want to Read" (TBR), and "Read" organization.
- **Goodreads/StoryGraph CSV Import:** Automated parsing tool to ingest years of reading history instantly.
- **CameraX Barcode Scanner:** Frictionless ISBN scanning pulling metadata directly from the Google Books API.
- **Offline-First Sync Matrix:** Last-Write-Wins (LWW) conflict resolution using the Wearable Data Layer API to sync seamlessly between local SQLite and cloud Firestore instances.

2. Mobile Command Center (Android)

- **Material You Dynamic Theming:** UI automatically adapts to system wallpaper colors with OLED-friendly "True Black" deep night modes equipped with amber/red text filters for late-night viewing.
- **Active Reading Hero Card:** A dominant ElevatedCard layout featuring the current book, percentage progress bar, and a quick-resume button.
- **Visual TBR Pipeline (Kanban):** A swipeable HorizontalPager separating books into "Backlog," "Shortlist," and "Up Next" to reduce decision fatigue.
- **Smart Batch-Logging:** A swipe-up gesture on dashboard items to instantly stack "+3 Chapters" or "+50 Pages" in a single action for fast-paced reading.

3. Wrist-First Interface (Wear OS)

- **Frictionless Session Logger:** A centered, large-text UI featuring native haptic feedback for "+1" and "+10" page increments, allowing for zero-look logging.
- **OLED Ambient Mode:** A battery-saving state that turns off background pixels and hides UI elements when the wrist is lowered, displaying only essential gray text.
- **Companion Care Timers:** Specialized, recurring timer presets on the Wear OS app for dual-tracking (e.g., a 15-minute countdown for an animal's outdoor or medication routine while simultaneously tracking pages read).
- **Complications System:** Circular progress rings for current book completion and daily page goal metrics on the main watch face.
- **End-of-Shift Low Power Sync:** Delays Firebase batch syncing until the watch is placed on its charging cradle to preserve battery life.

4. Advanced Analytics & Contextual Logging

- **Multi-Axis Qualitative Ratings:** Replaces standard 5-star systems with slider metrics for Pacing, Focus, and Vibe.
- **Granular DNF Tracking:** Captures exactly what percentage a book was abandoned and prompts for the specific reason (e.g., prose, pacing, characters).
- **Reading Velocity Analytics:** D3-style line charts tracking reading speed (pages per minute) across individual sessions and books.
- **Environment & Beverage Correlator:** Tags sessions with situational context, allowing the dashboard to correlate focus metrics against scenarios like morning coffee versus late-night whiskey neat.
- **The "Frequent Flyer" Streak Engine:** Anchors daily streak goals to a rolling 24-hour UTC block, preventing streak resets when crossing time zones for international conferences or long airport layovers.

5. Media & Format Adaptability

- **Serialized Content Mode:** Swaps standard "Pages" tracking for "Volumes" and "Chapters," optimized for tracking long-running manga runs.
- **The Adaptation Bridge (Manga to Screen):** Maps reading progress of serialized formats directly to their anime-canon television episodes, prompting users when they are cleared to watch specific episode blocks without spoilers.
- **Seamless TTS (Text-to-Speech) Handoff:** Integrates a native TTS engine for uploaded documents and EPUBs, allowing tired readers to transition seamlessly from visual reading to AI-generated audio without losing page-tracking progress.
- **Multi-Format Pacing:** Adjusts tracking metrics to "Hours & Minutes" for audiobooks.
- **Custom Whitepaper Tracking:** Allows manual creation of unpublished library items to track corporate documentation, restructuring proposals, or custom software manuals.

6. Ecosystem & Hardware Integrations

- **Stylus Quick-Capture:** Native hardware integration for devices supporting an active stylus, launching a floating scratchpad to handwrite and digitize margin notes via ML Kit Digital Ink Recognition.
- **Android Auto Module:** A simplified dashboard UI for logging audiobook progress or bookmarking timestamps directly from a vehicle's infotainment touchscreen.
- **Google Cast Big Screen Analytics:** Projects Kanban boards and yearly wrap-up heatmaps directly to connected TV streamer devices for large-scale visual planning.
- **ARM-Optimized Desktop PWA:** The desktop review portal is built strictly as a Progressive Web App with local caching, ensuring lightning-fast startup and minimal RAM usage on modern ARM-based Windows architecture, bypassing the need for heavy emulation.

7. Lifestyle & Shift-Aware Planning

- **Shift-Aware "Wind Down" Reminders:** Abandons static 9:00 PM pings in favor of dynamic anchors, allowing users working late shifts from 4:00 PM to 3:00 AM to schedule silent watch vibrations during their actual wind-down window.
- **The Transit Optimizer:** A time-boxed suggestion engine that cross-references historical reading velocity with the "Shortlist" board to suggest the exact novel, whitepaper, or manga volume that can be realistically finished during a specific transit window or airport layover.
- **The "Life Interruption" Pause:** A giant pause toggle on the watch and phone to accurately preserve reading speed metrics when stepping away for sudden distractions.
- **Location-Based Reading Zones:** Geofenced triggers that silently prompt reading sessions upon arrival at designated environments.
- **Guilt-Free Snooze:** Quick-action notifications allowing users to seamlessly push a daily reading goal to the next day without breaking their mathematical streak.

8. Journaling & Knowledge Management

- **Wrist Dictaphone:** A Wear OS microphone complication that converts speech to text, attaches the current page timestamp, and fires the note directly to Firestore.
- **Chronological Margin Notes:** Displays journal entries on a vertical timeline corresponding strictly to the currentPage integer.
- **Markdown Formatting:** Full technical markdown support in the editor for structuring code blocks and bullet points when reading technical architecture guides.
- **Dynamic Audio Intent Linking:** Binds specific streaming playlists to book profiles, automatically launching ambient soundscapes or alternative rock tracks when a session timer starts.